

MULTIMEDIA



UNIVERSITY

STUDENT ID NO

--	--	--	--	--	--	--	--	--	--

MULTIMEDIA UNIVERSITY

FINAL EXAMINATION

TRIMESTER 1, 2019/2020

TCN2141 – COMPUTER NETWORKS

(All sections / Groups)

17 OCTOBER 2019
2.30 p.m. – 4.30 p.m.
(2 Hours)

INSTRUCTIONS TO STUDENTS

1. This question paper consists of 4 pages with 5 questions only
2. Attempt **ALL FIVE** questions.
3. All questions carry equal marks and the distribution of the marks for each question is given.
4. Please print all your answers in the answer booklet provided.

QUESTION 1

- a) Briefly explain the **THREE** phases in the virtual-circuit approach to switching.
- b) Differentiate between multiple unicasting and multicasting. In which network service application, multicasting is used.
- c) Given a host IP address with CIDR as 179.5.30.5/18. The administrator wants to create 16 subnets:
 - i. Find the subnet mask required.
 - ii. What is the subnet number for the 4th subnet?
 - iii. What is the subnet broadcast address for 7th subnet?
 - iv. What are assignable addresses for the 13th subnet?
 - v. What is 10th usable IP address in 9th subnet?

[3 + 2 + 5 = 10 Marks]

QUESTION 2

- a) Based on Software-Defined Networking (SDN) paradigm.
 - i. Briefly explain the data plane concept and its key function(s).
 - ii. Briefly explain the control plane and its key function(s).
- b) Briefly describe **FOUR** general techniques to improve quality of service (QoS) in a network.
- c) Briefly describe the difference between group-shared tree and a source-based tree in the context of multicast routing.

[4 + 4 + 2 = 10 Marks]

Continued...

QUESTION 3

- a) Consider sending a 4850-byte datagram into a link that has an MTU datagram of 1500 bytes. Suppose the original datagram is stamped with the identification number 235411 as shown in table below (Original IP datagram). Use IP fragments table below to give your answer.

Original IP Datagram					
Sequence	Identifier	Total Length	DF	MF	Offset
0	112233	4850	0	0	0

IP Fragments (Ethernet)						Range	
Sequence	Identifier	Total Length	DF	MF	Offset	Start	End

- b) Compress the following IPv6 addresses

- i. 2001:0db8:0000:000b:0000:0000:0000:001A
- ii. 2008:1266:0000:0000:0AE4:0000:005B:06B0
- iii. 2E21:0000:0000:0000:0011:0002:00C3:00A4

- c) What is Autonomous System (AS)? List **THREE** types of autonomous systems (ASs)

[5+ 3 + 2 =10 Marks]

Continued...

QUESTION 4

- a) Briefly explain the purpose of an Address Resolution Protocol (ARP).
- b) Briefly describe Distributed Hash Table (DHT). Why DHT applied in structured peer-to-peer (P2P) network overlay?
- c) Briefly explain the reason for TCP is preferred protocol to run HTTP, FTP, SMTP and POP3 protocols rather than on UDP.
- d) Briefly define subnetting, and supernetting, How do the subnet mask and supernet mask differ from a default mask in classful addressing?
- e) Briefly describe NAT (Network Address Translation) and its functions

[2 +2 + 2+ 2+ 2 = 10 Marks]

QUESTION 5

- a) Calculate IP header checksum if an IP header is given as below:
0x45000028a2694000800600000a00582722c5459f
- b) Validated the above (a) checksum output at receiver's side and show that the IP header has no error.
- c) Briefly explain **FOUR** types of delays in a packet-switched network.
- d) The value of the total length field in an IPv4 datagram is 216, and the value of the header length field is **5**. How many bytes of data is the packet carrying?

[4+1+4+ 1= 10 Marks]

End of Page